Q1: 16 March (Shift 1) - Single Correct

Which of the following is Lindlar catalyst?

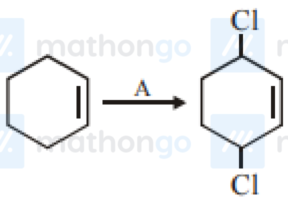
(1) Zinc chloride and HCl

(2) Cold dilute solution of $KMnO_4$

(3) Sodium and Liquid NH_3

(4) Partially deactivated palladised charcoal

Q2: 16 March (Shift 2) - Single Correct



Identify the reagent(s) 'A' and condition(s) for the reaction:

(1) $A = HCl$; Anhydrous $AlCl_3$

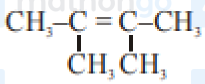
(2) $A = HCl, ZnCl_2$

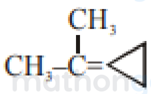
(3) $A = Cl_2$; UV light

(4) $A = Cl_2$; dark, Anhydrous $AlCl_3$

Q3: 16 March (Shift 2) - Single Correct

An unsaturated hydrocarbon X on ozonolysis gives A. Compound A when warmed with ammonical silver nitrate forms a bright silver mirror along the sides of the test tube. The unsaturated hydrocarbon X is:

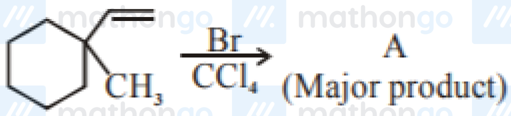
(1) 

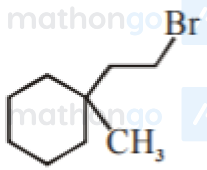
(2) 

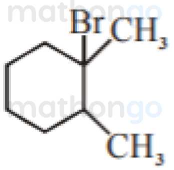
(3) $HC \equiv C - CH_2 - CH_3$

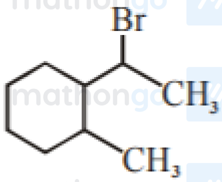
(4) $CH_3 - C \equiv C - CH_3$

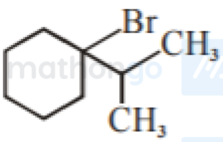
Q4: 17 March (Shift 1) - Single Correct



(1) 

(2) 

(3) 

(4) 

Q5: 17 March (Shift 2) - Single Correct

Given below are two statements:

Statement-I : 2 -methylbutane on oxidation with $KMnO_4$ gives 2 -methylbutan- 2 – o1.

Statement-II : n-alkanes can be easily oxidised to corresponding alcohol with $KMnO_4$. Choose the correct option :

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

Q6: 17 March (Shift 2) - Numerical

The total number of $C - C$ sigma bond/s in mesityl oxide $(C_6H_{10}O)$ is ____ (Round off to the Nearest Integer).

Answer Key

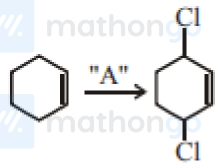
Q1 (4) **Q2** (3) **Q3** (3) **Q4** (4)

Q5 (3) **Q6** (5)

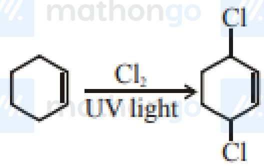
Q1 (4)

Partially deactivated palladised charcoal $(H_2/pd/CaCO_3)$ is lindlar catalyst.

Q2 (3)



For substitution at allylic position in the given compound, the reagent used is $Cl_2/$ uv light. The reaction is free radical halogenation.



Q3 (3)

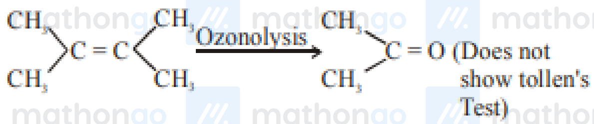
$$\underset{\text{Unsaturated Hydrocarbon}}{(X)} \xrightarrow{\text{Ozonolysis}} (A) \xrightarrow[\text{(Tollen's regent)}]{\text{Ammonical } AgNO_3} \underset{\text{silver mirror}}{Ag\downarrow}$$

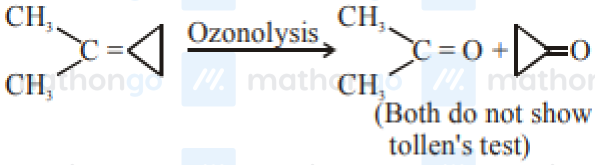
As (A) compound given positive tollen's test hence it may consist -CHO (aldehyde group). or it can be HCOOH So for the given option:

(c)

$$CH_3-CH_2-C\equiv CH \xrightarrow{\text{Ozonolysis}} CH_3CH_2COOH + \underset{\substack{(A) \\ \text{+ve tollen's} \\ \text{test}}}{HCOOH}$$

and for other compounds (options):

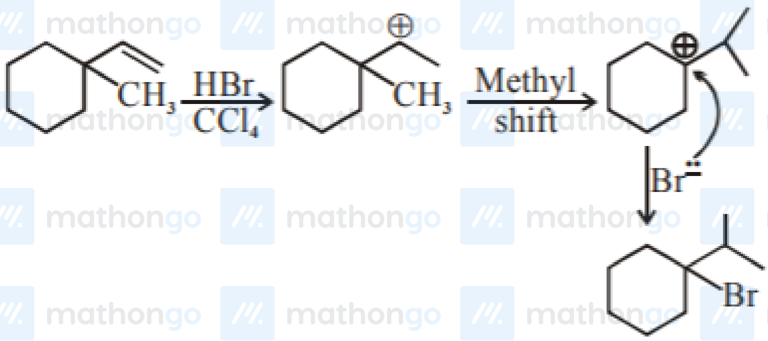
(a) 

(b) 

(d) $CH_3 - C \equiv C - CH_3 \xrightarrow{\text{Ozonolysis}} 2CH_3 - COOH$

(Does not show tollen's test)

Q4 (4)



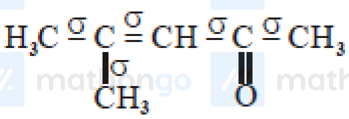
Q5 (3)

Alkane are very less reactive, tertiary hydrogen can oxidise to alcohal with $KMnO_4$

2-methyl-butane

Q6 (5)

Mesityle oxide



$\therefore \quad \mathrm{C}_\sigma\mathrm{C} = 5$